

Sergei Ermolaev

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📍 Nizhny Novgorod (UTC+3)

🇷🇺 RU CV

Summary

AI / Machine Learning Engineer with 3+ years of experience in ML, including **2.5 years** of end-to-end production ML systems. My main focus is building practical LLM-based systems: RAG, AI agents, LLM inference, hybrid search, evaluation, and production monitoring. Open to remote work and relocation.

Experience

Proton (IT company of AGAT Group, *top-5 Russian automotive dealer group* [🔗](#), 120+ showrooms)

Nizhny Novgorod
Nov 2023 – Present

Machine Learning Engineer:

- Built an end-to-end **multi-label call classification** pipeline: collected, cleaned, and analyzed the dataset, ran experiments with multiple models and frameworks, and improved sample-wise F1 from **79% to 94%** and EMR from **3% to 61.6%**; selected a CPU-efficient inference setup suitable for sub-second processing of 1000+ daily calls, supporting an estimated **~\$1.7M business impact**.
- Built a **production RAG assistant** using **RAGFlow + vLLM**; owned document ingestion, Docker deployment, monitoring, tracing, and quality evaluation, achieving **4.7/5 user satisfaction**, **95% faithfulness**, and **p99 TTFT of 4.6s** under load.
- Designed and implemented a ticket routing **AI agent** for an internal workflow system using RAG, Guardrails, Memory, LangGraph, LangFuse, and vLLM; owned the full development cycle, including **quality and performance monitoring**.
- Developed a car parts search system that reduced manual requests from **300 to 20/day**; implemented **hybrid search**, custom ranking logic, confidence thresholds, FastAPI service, Docker deployment, and production documentation.
- Built an **LLM-based call labeling** pipeline using Mistral/Qwen, zero-/few-shot prompting, and **LoRA Qwen2.5-3B**; achieved up to **90% accuracy** in experiments and deployed an XLinear baseline with 88% accuracy for CPU inference.
- Built a **vehicle test-drive tracking** pipeline using **YOLOv5**, multi-threaded video processing, and ROI cropping; improved tracking accuracy from **65% to 87%** and reached up to **15 FPS** on a consumer-grade GPU.

Projects

HSE University Nizhny Novgorod (Academic R&D)

Sept 2022 – May
2023

ML Engineer | *WhisperX, Llama.cpp, aiogram, guidance*. [GitHub](#) [🔗](#)

- Developed an autonomous AI assistant (Telegram Bot on aiogram) for audio transcription and summarization with a focus on data privacy. Designed an ASR pipeline based on WhisperX with text alignment and regex noise filtering. Implemented hybrid (CPU/GPU) inference of the quantized LLM via llama.cpp with custom memory management and strict generation control using the guidance framework. The system runs without external APIs and optimized for local execution on consumer hardware (from 12GB VRAM).

HSE University Nizhny Novgorod (Joint R&D project with business)

Feb 2021 – May 2021

ML Researcher | *Selenium, Colabeler, torchvision*

- Designed a data collection and preparation pipeline for a CV model detecting car damage. Collected the dataset, performed labeling, augmentation, and class balancing.

Education

HSE University

Master's degree in Data Mining

Nizhny Novgorod, Russia

Sept 2022 – June 2024

HSE University

Bachelor's degree in Applied Mathematics and Informatics

Nizhny Novgorod, Russia

Sept 2018 – June 2022

Professional Development

[Yandex School of Data Analysis \(SHAD\) Intensive: LLM Scaling Week](#) 

Nov 10, 2025 – Nov 14, 2025

[Yandex School of Data Analysis: Agents Week](#) 

Apr 6, 2026 – Apr 10, 2026

Skills

DL/ML: Python, PyTorch, vLLM, Transformers, LangChain, LangGraph, llama.cpp, DeepEval, RAGFlow, NumPy, prompt engineering, Infinity, Unsloth, scikit-learn, WhisperX, spaCy, Ultralytics

Infra/Tools: Linux, Docker, FastAPI, Git, DVC, Streamlit, LangFuse, W&B, matplotlib, Grafana, Kubeflow

Data: pandas, polars, SQL

Languages: Russian - Native, English - B2 / Upper-Intermediate